

Muhammad Usama

Senior AI Engineer | Generative AI | LLMs | RAG | MLOps

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SUMMARY

Senior AI Engineer with 6+ years of experience designing and deploying LLM and RAG-based systems across AWS, Azure, and GCP, supporting 50K+ automated decisions monthly. Built and scaled multi-agent AI platforms and Kubernetes-based inference systems handling 10K+ daily requests with 99.9% uptime, improving processing accuracy by up to 30%. Implemented MLOps pipelines and model deployment workflows, reducing regression issues by 25% and improving system reliability by 20–40%.

SKILLS

- AI & Generative AI:** Generative AI, Large Language Models (LLMs), Agentic AI, Multi-Agent Systems, Retrieval-Augmented Generation (RAG), Prompt Engineering, LLMOps, LangGraph, CrewAI
- Machine Learning & Data:** Machine Learning, Deep Learning, Natural Language Processing (NLP), Computer Vision, Feature Engineering, Model Evaluation, Model Deployment, YOLO, Mask R-CNN, Segment Anything Model (SAM), SadTalker, Wav2Lip
- Backend & Systems:** Python, FastAPI, Django, REST APIs, Microservices Architecture, Distributed Systems, Scalable AI Systems
- Cloud & Infrastructure:** Amazon Web Services (AWS), AWS Bedrock, AWS Lambda, EC2, S3, Microsoft Azure, Google Cloud Platform (GCP), Kubernetes (EKS, GKE, AKS), Docker, GPU Computing
- MLOps & DevOps:** MLOps, MLflow, DVC, CI/CD Pipelines, Jenkins, GitHub Actions, Infrastructure as Code (Terraform, CloudFormation)
- Data Engineering & Tools:** Apache Airflow, ETL Pipelines, Data Engineering, Vector Databases, Pinecone, FAISS, GraphQL

PROFESSIONAL EXPERIENCE

- Senior AI Engineer**
Folium AI

- Built AI avatar and video generation pipelines using **SadTalker**, **Wav2Lip**, and **Hunyuan** on A100 (80GB) GPUs, enabling scalable multi-modal content generation and reducing media production time by 50–60%.
 - Architected and deployed LLM and RAG-based systems across AWS, Azure, and GCP, supporting 50K+ automated decisions per month.
 - Designed multi-agent AI workflows using LangGraph and AWS Bedrock, improving document processing accuracy by 30% and reducing manual effort by 40%.
 - Built and scaled Kubernetes-based inference systems handling 10K+ daily requests with 99.9% uptime.
 - Developed hybrid RAG pipelines using FAISS and Pinecone, increasing answer accuracy from 70% to 88%.
 - Implemented MLOps pipelines using MLflow, DVC, and CI/CD tools, reducing model regression issues by 25%.
 - Engineered event-driven FastAPI and Django services processing millions of API requests per month with low latency.
 - Led technical design and mentored 6+ engineers, improving delivery efficiency and standardizing AI system architecture across projects.

06/2025 – Present
Remote
- Lead AI/ML engineer (Contract)**
Turing

- Led RLHF-based fine-tuning of Meta LLaMA models, improving response alignment with human preferences by 20% based on evaluation metrics.
 - Built and optimized reward models for preference learning, increasing response quality scores by 15% across validation datasets.
 - Curated and processed large-scale human feedback datasets, improving data quality and consistency for training pipelines.
 - Implemented policy optimization workflows, reducing unsafe or low-quality outputs by 18% during model evaluation.

01/2025 – 03/2025

- Developed scalable training and evaluation pipelines, reducing experimentation time by 25% and improving iteration speed.

Data Scientist

Devsinc

02/2024 – 06/2025
Remote

- Developed and deployed machine learning models for identity verification and fraud detection, improving prediction accuracy by 18% across large-scale datasets.
- Built data pipelines processing structured and unstructured data, reducing data processing time by 25% and improving pipeline reliability.
- Implemented feature engineering and model optimization techniques, increasing model performance metrics by 15–20%.
- Designed evaluation frameworks and monitoring systems, reducing model drift impact by 20% and improving consistency in production.
- Collaborated with cross-functional teams to integrate ML models into production systems, supporting real-time decision-making workflows at scale.

Data Scientist (AI Developer → Data Scientist)

AiforMax

01/2020 – 07/2024
USA

- Progressed from AI Developer to Data Scientist within 4 years, leading development of fraud detection and identity verification systems on large-scale datasets.
- Built AML and KYC pipelines processing high-volume transaction and identity data, improving fraud detection recall by 15% and reducing false positives by 10%.
- Deployed CV/OCR models (YOLO, Mask R-CNN, custom OCR) for document verification, achieving 92%+ extraction accuracy across onboarding workflows.
- Optimized GPU-based training and inference pipelines for identity verification and liveness detection, reducing processing time by 30% using multi-GPU and CUDA tuning.
- Developed real-time compliance dashboards integrated with banking systems, supporting 5+ financial institutions with automated alerts and audit tracking.
- Engineered data pipelines and model workflows for large-scale processing, improving system throughput and enabling consistent real-time KYC monitoring.

 EDUCATION

BSCS, Information Technology

University Of Central Punjab
Grade: 3.73 CGPA

02/2020 – 03/2024
Lahore

 PROJECTS

AI-Powered Digital Credentialing Platform (VerixPro)

- Fine-tuned Mask R-CNN for blueprint segmentation to detect and label 6 object types, deploying dedicated EC2 models and building a scalable microservice inference pipeline with Django, Auto Scaling Groups, S3-based PDF processing, and distributed session management.
- Engineered an AI and blockchain-based credentialing platform with automated certificate generation and verification workflows, reducing fraud risk and increasing issuance efficiency while driving up to 80% engagement.

AI-Driven Environmental Risk Intelligence Platform (AreaHub.ai)

- Built Django-based microservices platform with an AI chatbot converting natural language queries into SQL for real-time property search, reducing response latency by 40% using async processing, caching, and streaming.
- Developed agent-based AI pipelines for listing creation and automated video generation (MoviePy, TTS), reducing manual effort by 70%, improving listing quality by 30%+, and increasing user engagement by 40%.

AI-Powered Student Attendance & Intervention Platform (Circle2Learn)

- Developed an AI-driven attendance management platform integrating with SIS systems to identify at-risk students and generate intervention plans, reducing report generation time from hours to under 5 minutes.
- Built data pipelines and analytics workflows to track attendance trends and automate decision support, improving early risk detection across large student populations and enabling real-time intervention strategies.

 CERTIFICATES

- DevOps Certification – DEVTOWN (MLSA & GDSC Collaboration)
- Clash of coders coding competition Issuing organization

 LANGUAGES

- English
- Urdu
- Punjabi